

Milestone 1.4: Synthesis: Three Lenses, One Problem

INFO 521 · Machine Learning Foundations · project milestone brief ·
One Problem, Three Lenses · Synthesis

How to read this

This is the assignment brief for one project milestone. The Satisfactory criteria below are authoritative: every one must be met, and it is all-or-nothing rather than partial credit. A Not-Yet returns with targeted feedback and one revise cycle. Do the work in the notebook distributed through Classroom 50; this PDF is the brief, not the workspace.

Course-schedule placement. Part 1 of the project: due Week 5, the midpoint. This milestone covers M1 to M4 and is due Week 5. Gated by structured peer review.

The payoff

You have answered the same clinical question three ways. Now make the comparison *visible* and *argued*. This synthesis is where the conceptual connections become concrete, the same prediction, seen as a point, a likelihood, and a posterior.

What “Satisfactory” means for 1.4

- ☐ A **single comparison panel** shows the same age $\rightarrow$ SBP prediction under all three lenses: OLS point estimate, MLE with $\pm 2\hat{\sigma}$ band, and posterior-predictive credible interval.
- ☐ A written **compare-and-contrast** (≈ 400 – 600 words) of the three approaches: what each assumes, what each produces, how the parameter estimates relate, and when you would reach for each.
- ☐ The notebook **runs end-to-end** against the pinned environment.
- ☐ Structured **peer review** completed (rubric posted on D2L).

The comparison panel

```
import numpy as np
from info521 import data, plotting
plotting.use_course_style()
ds = data.load_clinical()
age = data.primary_predictor(ds)
y = ds["y"]

# TODO: bring in your fits from 1.1-1.3 (import your module or re-fit) and draw
# ONE figure overlaying: OLS line, MLE mean + error band, posterior mean +
# credible band, over the data scatter. Use distinct Okabe-Ito colors and a
# legend.
```

The argument

Address, in prose:

1. **Assumptions.** What does each lens assume about the world (about error, about the parameters)?
2. **Outputs.** A number, a likelihood surface, a distribution, what do you actually get back, and what can you say with each?
3. **Relationships.** Why does the MLE for w coincide with OLS? Under what conditions does the posterior mean approach the MLE?
4. **Choice.** Given a new clinical question, which lens would you pick, and why?

Reflection (1 paragraph)

What did switching lenses teach you that any single lens would have hidden?

* Bridge to Part 2

Everything here rested on a **conjugate** prior–likelihood pair, which kept the posterior in closed form. In Part 2 you binarize systolic BP into a **hypertension** label (the ACC/AHA rule: $SBP \geq 130$ or $DBP \geq 80$) and predict it from the **non-BP** features: the blood-pressure columns are held out so the label can't leak into its own predictors. Conjugacy breaks, the posterior goes intractable, and you meet Newton–Raphson, the Laplace approximation, and MCMC, then SVMs, clustering, and PCA across the *full* (non-BP) feature set, not just age.